Hardik Hemant Pandey

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Portfolio | GitHub | LinkedIn

Research interests: computer networking and distributed systems, including protocol measurement, testbed design, and empirical evaluation of network transport protocols.

Education

University of Glasgow, Glasgow, UK — Sept 2025 – Sept 2026
MSc Computing Science

Relevant coursework: Programming and Systems Development; Secure Systems Programming in Rust; Research and Professional Skills.

Chandigarh University, India — 2021 – 2025
B.E. Computer Science & Engineering — CGPA: 8.14/10

Relevant coursework: Computer Networks; Operating Systems; Cloud Computing & Distributed Systems; Data Structures & Algorithms

Research Experience

MSc Dissertation: A Reproducible Testbed for Media over QUIC Transport: Interop, Controlled Impairments and Baselines

Supervisor: Prof. Colin Perkins | Code: github.com/Hardik-7892/moqt-testbed — 2026

- Built a ContainerNet/Docker testbed to evaluate three MOQT implementations (moq-rs, imquic, moxygen) for interoperability and delivery under basic, chain and fan-out topologies.
- Added wire level checks for draft support using ALPN handshake tiers, and for delivery using per-subscriber byte hash verification.
- Found a fan-out bug in a production relay that logged all subscribers as healthy but forwarded data to only one, confirmed as version specific against an earlier pin of the same code.
- Applied per-link loss, delay, bandwidth and queue limits with Linux `tc`, and benchmarked MOQT against LL-DASH over HTTP/2 and HTTP/3, with $\sim$37% gain of H3 over H2 and MOQT at $\sim$1.95$\times$ same content H3 delivery.
- Added full reproducibility with pinned commits, YAML test plans and auto pass/partial/fail verdicts.

Relevant Experience

Samsung Electro-Mechanics, Bangalore, India — Oct 2024 – Jun 2025
Student Intern, Kubernetes & ML Systems

- Deployed and managed Kubernetes clusters supporting multi-team, multi-node distributed ML workflows.
- Monitored distributed system performance with Grafana and Prometheus, identifying and resolving pipeline bottlenecks across development and production environments.

Publications

Pandey, H. H. *et al.* **"Advance Approaches in AI-Powered Website Development for Educational Delivery via Ed-Tech."** *2024 International Conference on Trends in Quantum Computing and Emerging Business Technologies (TQCEBT)*, IEEE, 2024. DOI: 10.1109/TQCEBT59414.2024.10545211. *First author (undergraduate).*

Technical Skills

- **Languages:** Python, C/C++, Rust, Java
- **Systems & Networking:** Linux, Docker, Containernet/Mininet, QUIC / HTTP-3, TCP/IP, Wireshark, `tc` (Linux traffic control), Git
- **Infrastructure & Monitoring:** Kubernetes, Grafana, Prometheus
- **Miscellaneous:** LaTeX, Zotero